

Multiple Choice (10 marks)

1. In Python 3, what is `['a', 'Chemistry', 0, 1][-3]`?
 - (a) a
 - (b) Chemistry
 - (c) 0
 - (d) 1
2. Which of the following instruction-set features is NOT generally considered an obstacle to aggressive pipelining of an integer unit?
 - (a) Condition codes set by every instruction
 - (b) Variable-length encoding of instructions
 - (c) Instructions requiring widely varying numbers of cycles to execute
 - (d) Several different classes (sets) of registers
3. In Python 3, which of the following function removes all leading and trailing whitespace in string?
 - (a) `replace(old, new [, max])`
 - (b) `strip([chars])`
 - (c) `swapcase()`
 - (d) `title()`
4. In Python 3, what is the following function returns item from the list with max value?
 - (a) `cmp(list)`
 - (b) `len(list)`
 - (c) `max(list)`
 - (d) `min(list)`
5. Which of the following best describes the primary way in which a distributed denial-ofservice (D D o S) attack differs from a denial-ofservice (D o S) attack?
 - (a) The goal of the attack
 - (b) The number of computers being attacked
 - (c) The number of computers launching the attack
 - (d) The time period in which the attack occurs

True / False (10 marks)

Answer True or False

6. True or False: The minimum value of the list $l = [1, 2, 3, 4]$ in Python 3 is 1.
7. True or False: The function $O(N)$ grows slower than $O(N^{1/N})$ as N approaches infinity.
8. True or False: Bitmap graphics can represent realistic lighting and shading effects accurately through pixel manipulation.
9. True or False: In a complete K -ary tree of depth N , the ratio of nonterminal nodes to total nodes approximates to $(K-1)/K$.
10. True or False: A 3-way, set-associative cache allows each main memory word to be stored in any of 3 specific cache locations.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a python function to count the number of rectangles in a circle of radius r .
12. Write a python function to find even numbers from a mixed list.

End of Paper